

Differentiation of Implicit Functions

Section 2.6

Recall from the previous sections, that when we take the derivative of $y = f(x)$ then $y' = f'(x)$ where $f(x)$ is a function in terms of x (i.e., the only variable in the function is x).

Example: If $y = x^3 + 2x$ then $y' = \frac{dy}{dx} = 3x^2 + 2$

So, in other words, to take a derivative this way, we have to have the equation solved for “y.”

Example: If $x = x^3 + 2x - y$. Here we first have to solve for y . So $y = x^3 + 2x - x \Rightarrow y = x^3 + x$, then $y' = \frac{dy}{dx} = 3x^2 + 1$.

Example: If $yx = x^3 + 2xy^2 - y$. Again, we have to solve for y in order to take a derivative in the way that we have learned in the preceding chapters. However, (as you can see in this case) it is not always easy/possible to do so.

HENCE: Implicit Differentiation!

A. Implicit Differentiation

Implicit Differentiation: Differentiation of a function where one variable (typically y) is not explicitly expressed as a function of another variable (typically x).

Here's how it works.

- It is important to pay attention to the notation. If we are given an equation in terms of x and y , and asked to find y' or $\frac{dy}{dx}$, we need to see that we are finding the derivative of y , with respect to x .
- We will treat both x and y like a variable, and take derivatives of each, but;
- When we take a derivative of a term containing “ x ” we will proceed as usual.
- When we take a derivative of a term containing “ y ” we will proceed as usual AND then also multiply the derivative of that term by $\frac{dy}{dx}$ (or y').
- We will use product, quotient, and chain rules as needed
- After differentiating, solve for (i.e., isolate) y' or $\frac{dy}{dx}$.

Example A: Find $\frac{dy}{dx}$ for $x = x^3 + y^2$.

$$\frac{d(x)}{dx} = \frac{d(x^3)}{dx} + \frac{d(y^2)}{dx} \Rightarrow 1 = 3x^2 + 2y \cdot \frac{dy}{dx}$$

Since we are trying to find $\frac{dy}{dx}$, isolate $\frac{dy}{dx}$ in our equation: $1 - 3x^2 = 2y \cdot \frac{dy}{dx} \Rightarrow \frac{1 - 3x^2}{2y} = \frac{dy}{dx}$

Example B: Find $\frac{dy}{dx}$ for $x = 4x^3 + y^2 - 8y$.

$$\frac{d(x)}{dx} = \frac{d(4x^3)}{dx} + \frac{d(y^2)}{dx} - \frac{d(8y)}{dx} \Rightarrow 1 = 12x^2 + 2y \cdot \frac{dy}{dx} - 8 \cdot \frac{dy}{dx}$$

Since we are trying to find $\frac{dy}{dx}$, isolate $\frac{dy}{dx}$ in our equation:

$$1 - 12x^2 = 2y \cdot \frac{dy}{dx} - 8 \cdot \frac{dy}{dx} \Rightarrow 1 - 12x^2 = \frac{dy}{dx}(2y - 8) \Rightarrow \frac{1 - 12x^2}{2y - 8} = \frac{dy}{dx}$$

Example C: Find y' for $x = x^3y^2 - 3y^3$. (Notice that in this problem we have x^3y^2 —a product of x and y . Here we will have to use the product rule.)

$$\frac{d(x)}{dx} = \frac{d(x^3y^2)}{dx} - \frac{d(3y^3)}{dx} \Rightarrow 1 = [(3x^2)(y^2) + (x^3)(2y \cdot y')] - 9y^2 \cdot y'$$

$$\Rightarrow 1 - (3x^2)(y^2) = (x^3 \cdot 2y) \cdot y' - (9y^2) \cdot y' \Rightarrow 1 - (3x^2)(y^2) = y'(x^3 \cdot 2y - 9y^2)$$

$$\Rightarrow \frac{1 - 3x^2y^2}{2x^3y - 9y^2} = y'$$

Examples: Find $\frac{dy}{dx}$ for the following:

1. (video) $x^2 + y - 11 = 8x$

2. (video) $x^2 + 3y^2 - 4x + 7y = 14$

3. (video) $x^3y^2 + 9y + 3x = 4$

4. (video) Find y' for $3y + 12 - x^2 - y^4 = 0$ and evaluate at $(2, -1)$

5. $x^2 + 2y^2 - 11 = 0$

6. $y^2x - \frac{5y}{x+1} + 3x = 4$

7. Find y' for $2y + 5 - x^2 - y^3 = 0$ and evaluate at $(2, -1)$

8. Find $\frac{dA}{dt}$ for $A = \pi r^2$

9. Find $\frac{dV}{dt}$ for $V = \frac{1}{3}\pi r^2 h$

10. If $f(x) + x^7 \cdot [f(x)]^3 = 11$ and $f(2) = 6$, find $f'(2) =$

11. Use implicit differentiation to find an equation for the tangent line to the curve $4x^2 - 4xy - y^3 = 84$ at the point $(1, -4)$.